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Software Requirements Specification

Multi-Tenant Recruiter Vetting & Profile Matching Pipeline (v3.2)

Architecture: LangGraph + Dynamic In-Context Matching — Storage: Stateless Session State — Budget: \$0.00/mo

Executive Summary: Multi-Tenant Architecture & Capabilities

How Multi-User Support Works (Stateless Privacy-By-Design):

- Dynamic Resume Onboarding:** Any user uploads their PDF resume in the UI. The system parses the document on the fly using `pdfplumber` and converts it into a standardized JSON Schema (`skills`, `experience`, `domain_constraints`).
- In-Memory Session Storage:** Parsed profile JSONs are stored exclusively in Streamlit session memory (`st.session_state`). ****Zero user data is persisted to disk or external databases****, enabling concurrent multi-tenant usage at \$0 infrastructure cost.
- 3-Tier Agent Pipeline:** When a message is submitted, the system triggers a ****LangGraph state machine****:
 - Tier 1 (Rule Filtering):** Fast regex checks reject spam and domain mismatches in ~50ms.
 - Tier 2 (Tavily Search API):** Queries web data to verify company legitimacy and funding, abstaining on unindexed stealth startups (`ABSTAIN_NEEDS_HUMAN`).
 - Tier 3 (In-Context Schema Matching):** Dynamically injects the user's session profile JSON alongside extracted job requirements into the LLM prompt to compute an objective fit score.

1. System Overview & Architecture Revisions

1.1 Purpose

This document specifies software requirements for v3.2 of the Autonomous Recruiter Opportunity Vetting Agent. Version 3.2 introduces dynamic multi-tenant user handling, moving away from hardcoded candidate resumes toward on-the-fly PDF parsing, schema normalization, and in-context matching.

1.2 Key Architectural Upgrades (v3.2)

- Dynamic PDF Resume Parsing:** Integrated `pdfplumber` + fast LLM extraction to convert user-uploaded PDFs into structured JSON profiles upon session initialization.
- Stateless Multi-Tenant Isolation:** Candidate profiles exist strictly within active browser session memory (`st.session_state`), ensuring total privacy compliance and zero database overhead.
- In-Context Matching Strategy:** Replaced vector chunking with ****Structured JSON In-Context Injection**** for Tier-3 candidate alignment, eliminating embedding latency while preserving full resume context.
- Tavily Search API Wrapper:** Utilizes Tavily's dedicated LLM search API (1,000 free queries/mo) to prevent IP rate-limiting on shared Hugging Face container environments.

2. End-to-End System Workflow Architecture



3. Technical Requirements & Infrastructure Stack

Component	Technology Selection	Free-Tier Specification & Functionality
User Interface	Streamlit (Python)	Web dashboard with file uploaders, state visualization, and model telemetry.
Cloud Hosting	Hugging Face Spaces	Free CPU Basic container (2 vCPU, 16 GB RAM) hosting the app runtime.
State Machine	LangGraph	Manages cyclic graph execution, retry rules, and fallback node transitions.
PDF Parser	pdfplumber	Lightweight Python library for extracting clean raw text from uploaded resume PDFs.
Session Memory	Streamlit Session State	In-memory key-value store holding the user's parsed profile JSON during active session.
Search API	Tavily Search API	Dedicated search API (1,000 free queries/mo) for verifying companies and funding.
Primary LLM	Groq API (Llama 3.3 70B)	Ultra-fast primary engine for schema extraction, search planning, and fit evaluation.
Fallback LLM	Gemini API (2.5 Flash)	Automated fallback endpoint triggered via with_fallbacks() during Groq HTTP 429 limits.
Observability	LangSmith API	Traces node execution latencies, token consumption, and graph routing paths.

4. Functional Requirements

- **FR-1 (Dynamic Resume Ingestion):** The system shall accept user-uploaded PDF resumes via Streamlit, extract raw text using `pdfplumber`, and format the output into a standardized JSON schema.
- **FR-2 (Session Memory Isolation):** Parsed profile JSON objects shall be stored in `st.session_state["candidate_profile"]` and automatically purged upon session termination or browser tab closure.
- **FR-3 (Tier-1 Heuristic Triage):** The pipeline shall evaluate sender domains and spam keywords in $\sim 50\text{ms}$, short-circuiting obvious spam at \$0 compute cost.
- **FR-4 (Tier-2 Tavily Web Investigation):** The agent shall construct structured web queries via Tavily API to verify company domain legitimacy, recent funding, and engineering focus.
- **FR-5 (Self-Correction Loop):** If search results are off-topic or noisy, the self-diagnosis node shall rewrite search parameters and retry up to two times.
- **FR-6 (Agentic Abstention):** The system shall emit an `ABSTAIN_NEEDS_HUMAN` classification when evaluating unindexed stealth startups to prevent hallucinated entity details.
- **FR-7 (Tier-3 Dynamic In-Context Matching):** The pipeline shall pass the session's active profile JSON alongside extracted role requirements into the LLM context window to compute fit scores (`HIGH_FIT` vs. `LOW_FIT`).
- **FR-8 (Multi-Model Failover):** If Groq API throws an HTTP 429 rate limit exception, execution shall automatically shift to Gemini 2.5 Flash without loss of state.
- **FR-9 (Telemetry & Evals):** Every execution run shall log node latency traces to LangSmith and remain testable via an offline `pytest` benchmark suite.

5. Target Performance Metrics & Failure Modes

5.1 Latency, Cost, and Token Budget Breakdown

Pipeline Stage	Avg Latency	Search Cost	Token Cost	Architectural Function
Resume PDF Parsing (Once)	$\sim 1.2\text{ s}$	\$0.00	~ 600 tokens	Converts raw PDF into structured JSON profile schema.
Tier 1: Heuristic Filter	$\sim 50\text{ ms}$	\$0.00	0 tokens	Drops domain mismatches and generic spam immediately.
Tier 2: Tavily Search API	$\sim 1.2\text{ s}$	1 API Credit	~ 350 tokens	Verifies entity existence, funding, and tech stack.
Tier 3: In-Context Match	$\sim 1.4\text{ s}$	\$0.00	~ 900 tokens	Evaluates job requirements against active profile JSON.
Total Per-Message Run	$\sim 2.65\text{ s}$	1 API Credit	$\sim 1,250$ tokens	Delivers full verdict with complete telemetry trace.

5.2 Documented Failure Modes

1. **Stealth AI Startup Over-Abstention:** High-value stealth startup pitches lacking indexed web presence will cause Tier 2 to trigger `ABSTAIN_NEEDS_HUMAN`. *Trade-off: Favors zero hallucination over false automated confidence.*
2. **Agency Recruiter Ambiguity:** Third-party recruiters who withhold client company names prevent web verification, resulting in human escalation.
3. **Complex PDF Layout Parsing:** Multi-column or heavily formatted PDF resumes may produce noisy text extraction from `pdfplumber`, requiring a secondary re-formatting prompt pass.